

CUSTOMER SEGMENTATION ANALYSIS

Driving Targeted Marketing & Personalized Experience



BrightNest

Home & Decor LTD

Data-Driven Customer Segmentation Report

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1. Executive Summary

This project presents a comprehensive customer segmentation analysis based on a publicly available dataset for “HomeNest Home & Decor LTD” company. The dataset contains transactional records from a business that sells a wide variety of consumer goods, including home décor, stationery, seasonal items and giftware.

The primary objective of this analysis was to identify distinct customer groups using robust clustering techniques. After data cleaning and preparation, essential metrics such as Recency, Frequency, Monetary Value and Average Purchase Size were engineered for each unique customer. Using K-Means clustering, four unique customer segments were derived, each representing different behavioural profiles within the customer base.

This segmentation provides a strategic lens into how customers interact with the business, from high-value loyal patrons to infrequent budget shoppers. These insights form the basis for targeted marketing, personalized engagement strategies and optimized resource allocation across sales and operations teams.

Business Impact -

This customer segmentation equips leadership with a data-driven foundation to make high-value decisions in marketing, inventory, customer retention and revenue forecasting. By understanding how different customer groups contribute to profitability, the organization can:

- Prioritize high-impact segments for revenue maximization
- Improve customer lifetime value through personalized campaigns
- Reduce churn risk through proactive engagement
- Align inventory and product strategy with behavioural patterns
- Enhance operational efficiency through better demand planning

2. Introduction

Customer segmentation plays a critical role in helping businesses understand the diversity within their buyer base and tailor strategies accordingly. For this project, a publicly available dataset for “HomeNest Home & Decor LTD” was used as the foundation for analyzing customer behaviour and identifying meaningful customer groups. The dataset contains two years of transaction-level information, including product codes, quantities purchased, invoice details, pricing and customer identifiers, offering rich visibility into how individuals interact with the business.

Before applying analytical techniques, the data underwent extensive preparation to remove inconsistencies, incorrect entries and incomplete records. Once cleaned, the dataset allowed for the creation of essential customer metrics that summarize purchasing behaviour at an individual level. These metrics form the basis of the segmentation model and provide a reliable representation of how frequently customers buy, how recently they engaged and the overall financial value they generate.

Using these behavioural indicators, customers were grouped through machine-learning-based clustering. This approach reveals natural patterns that would not be visible through manual observation alone. The resulting segments serve as a foundation for more effective customer management, allowing the business to develop differentiated retention plans, marketing initiatives and value-enhancement strategies backed by data.

3. Data Overview

The dataset contains two years of operational sales records for a UK-based wholesale and retail distributor. It captures granular invoice-level information that enables detailed reconstruction of customer purchasing behavior.

Dataset Composition -

The raw dataset includes 541,909 transaction rows across 8 key variables:

- InvoiceNo – Unique identifier for each transaction or credit note.
- StockCode – Product-level reference code for each item.
- Description – Text description of the purchased product.
- Quantity – Number of units ordered.
- InvoiceDate – Timestamp indicating when the transaction was processed.
- UnitPrice – Selling price per unit of product (in £).
- CustomerID – Unique numeric identifier assigned to each customer.
- Country – Country of the purchasing customer.

Data Characteristics -

The dataset reflects typical challenges found in real-world retail data:

- Missing Customer IDs on several transactions, requiring removal for customer-level aggregation.
- Negative quantities and invoices, representing product returns and cancellations.
- Duplicate descriptions and inconsistent text formatting, common in operational systems.
- Outliers in spending and quantity, driven by bulk orders from wholesale clients.

Data Preparation for Modelling -

To enable accurate customer clustering, the dataset was transformed into a customer-level table through the creation of RFM-based features:

- Recency – Days since the customer's last purchase.
- Frequency – Count of completed purchase invoices (excluding returns).
- Monetary – Total revenue generated by the customer.
- Average Purchase Value – Monetary divided by number of purchases.

Prior to modelling, the dataset underwent the following steps -

- Removal of invalid, missing or return-only customer records
- Treatment of infinite or erroneous values
- Standardization of numerical variables
- Preparation of a modelling-ready dataset for K-Means clustering

4. Exploratory Data Analysis (EDA)

A preliminary exploration of the raw transactional dataset was conducted to understand data structure, assess quality and identify key behavioural patterns prior to segmentation. The analysis focused on customer activity, product performance, temporal sales trends and country-level revenue contribution.

1. Dataset Structure & Basic Metrics

The dataset contains 168,631 transaction records across 2,997 unique customers and 2,720 unique products, representing a comprehensive view of retail purchasing behaviour. Key aggregate metrics include:

- Total Sales: £3.87M
- Average Order Value (AOV): £486.81
- Number of Orders: 7,961
- Average Sales per Transaction: £22.98

These metrics indicate a high-volume, low-ticket retail business model with strong repeat purchase potential.

Basic Dataset Metrics	
Metric	Values
Total Rows	168631
Unique Customers	2997
Unique Products	2720
Total Sales	3875487.99
Average Order Value (AOV)	486.8091936
No of Orders	7961
Average Sales	22.98206136

Fig 1. Pivot Table for Basic Dataset Metrics

2. Data Quality Check

A structured quality assessment identified the following:

- CustomerID: ~80% missing, indicating many anonymous or guest checkouts (expected in e-commerce retail).
- Product Description: ~0.86% missing.
- All other fields (InvoiceNo, Quantity, UnitPrice, InvoiceDate, Country) had no missing values.

Handling missing CustomerIDs was essential before clustering, as segmentation requires customer-level aggregation.

Data Quality Check		
Missing Value Summary		
Metric	Missing Count	Missing %
InvoiceNo	0	0
StockCode	0	0
Description	1454	0.862232554
Quantity	0	0
InvoiceDate	0	0
UnitPrice	0	0
CustomerID	135080	80.10342047
Country	0	0
Sales	0	0

Fig 2. Pivot Table for Data Quality Check

3. Product-Level Insights

The product analysis revealed a highly skewed demand distribution. The top-selling item, “PAPER CRAFT, LITTLE BIRDIE”, sold 80,895 units, far exceeding all other products.

Common themes among top sellers:

- Low-cost novelty items
- Decorative home accessories
- Impulse-purchase gift products

This pattern reinforces the business's positioning in the low-price, high-volume giftware segment.

Top 10 Products		
Rank	Description	Quantities Sold
1	PAPER CRAFT , LITTLE BIRDIE	80995
2	FAIRY CAKE FLANNEL ASSORTED COLOUR	3114
3	SMALL CHINESE STYLE SCISSOR	3000
4	WORLD WAR 2 GLIDERS ASSTD DESIGNS	2880
5	GROW A FLYTRAP OR SUNFLOWER IN TIN	2400
6	RED HARMONICA IN BOX	2100
7	GIN + TONIC DIET METAL SIGN	2000
8	WHITE HANGING HEART T-LIGHT HOLDER	1930
9	POPCORN HOLDER	1900
10	ASSORTED COLOUR T-LIGHT HOLDER	1824

Fig 3. Pivot Table for Top 10 Products

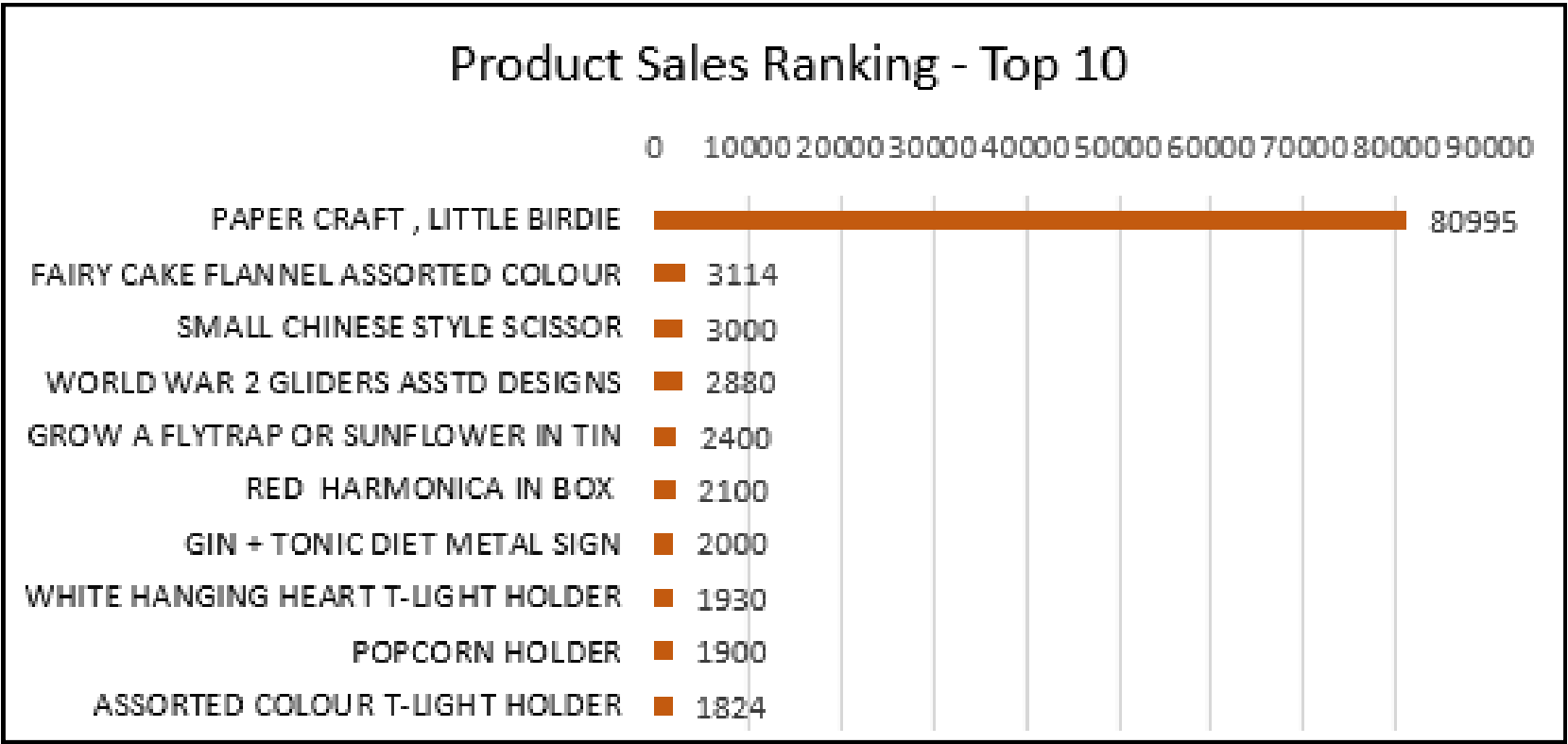


Fig 4. Bar Chart for Top 10 Products

4. Monthly Sales Performance

A monthly time-series analysis highlighted:

- Gradual growth throughout 2010 and early 2011
- A strong surge from mid-2011, peaking in November 2011
- A subsequent decline in December, likely seasonal or due to incomplete December data

Seasonality is clearly present, with sales strengthening towards Q4, typical for gift and home décor categories.

Monthly Sales	
Months ▼	Total Sales (£)
2010-01	46376.49
2010-02	47316.53
2010-03	23921.71
2010-05	31771.6
2010-06	31215.64
2010-07	53795.31
2010-08	39248.82
2010-09	38231.9
2010-10	33650.28
2010-12	17305.77
2011-01	239994.38
2011-02	209542.22
2011-03	261508.44
2011-04	329347.22
2011-05	322262.21
2011-06	305544.2
2011-07	322993.67
2011-08	257498.63
2011-09	421984.18
2011-10	291455.9
2011-11	345332.09
2011-12	205190.8
Grand Total	3875487.99

Fig 5. Pivot Table for Monthly Sales

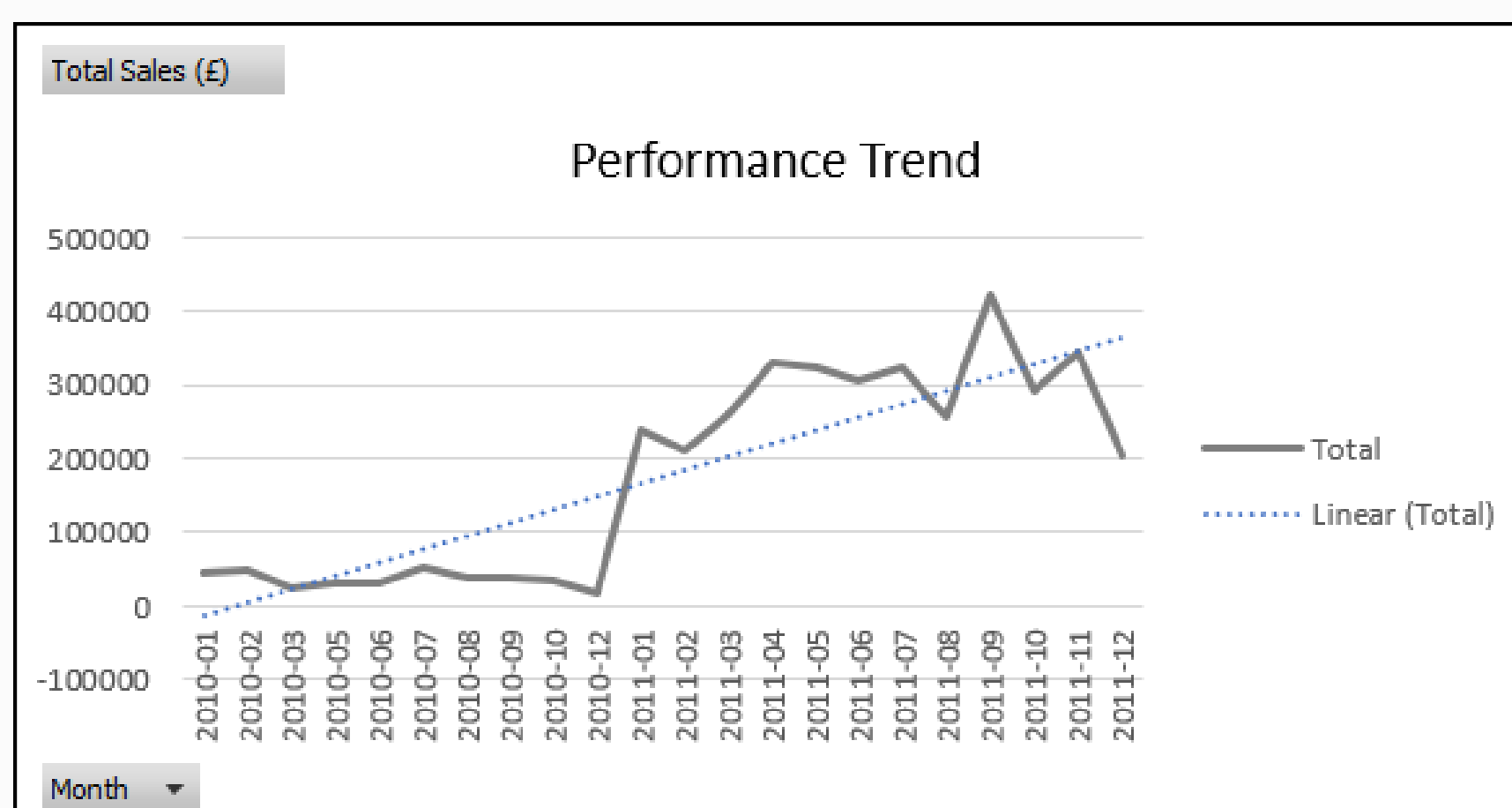


Fig 6. Performance Trend

5. Country-Level Revenue Analysis

The business is heavily UK-centric, with:

- United Kingdom contributing over £3.27M (~84% of total sales)
- Minimal contributions from other countries
- Strong secondary markets in EIRE, Germany, Netherlands and France

This concentration indicates significant opportunities for international expansion and targeted marketing outside the UK.

Country-Level EDA	
Country	Total Sales (£)
United Kingdom	3272810.14
EIRE	97500.23
Germany	94022.48
Netherlands	89850.96
France	85119.95
Australia	49209.71
Switzerland	27692.51
Spain	25007.25
Japan	17134.16
Belgium	16942.55
Norway	14655.1
Portugal	13162.27
Finland	11357.6
Sweden	9316.36
Singapore	8209.58
Channel Islands	7894.96
Cyprus	5385.91
Denmark	4863.56
Poland	3904.21
Italy	3664.11
Austria	2999.69
Iceland	2540.29
USA	2464.75
Greece	2099.28
Canada	1751.88
Lithuania	1661.06
Unspecified	1151.78
United Arab Emira	975.54
Malta	833.16
Bahrain	459.4
European Commu	432.05
Czech Republic	277.48
Israel	138.03
Grand Total	3875487.99

Fig 7. Pivot Table for
Country-Level EDA

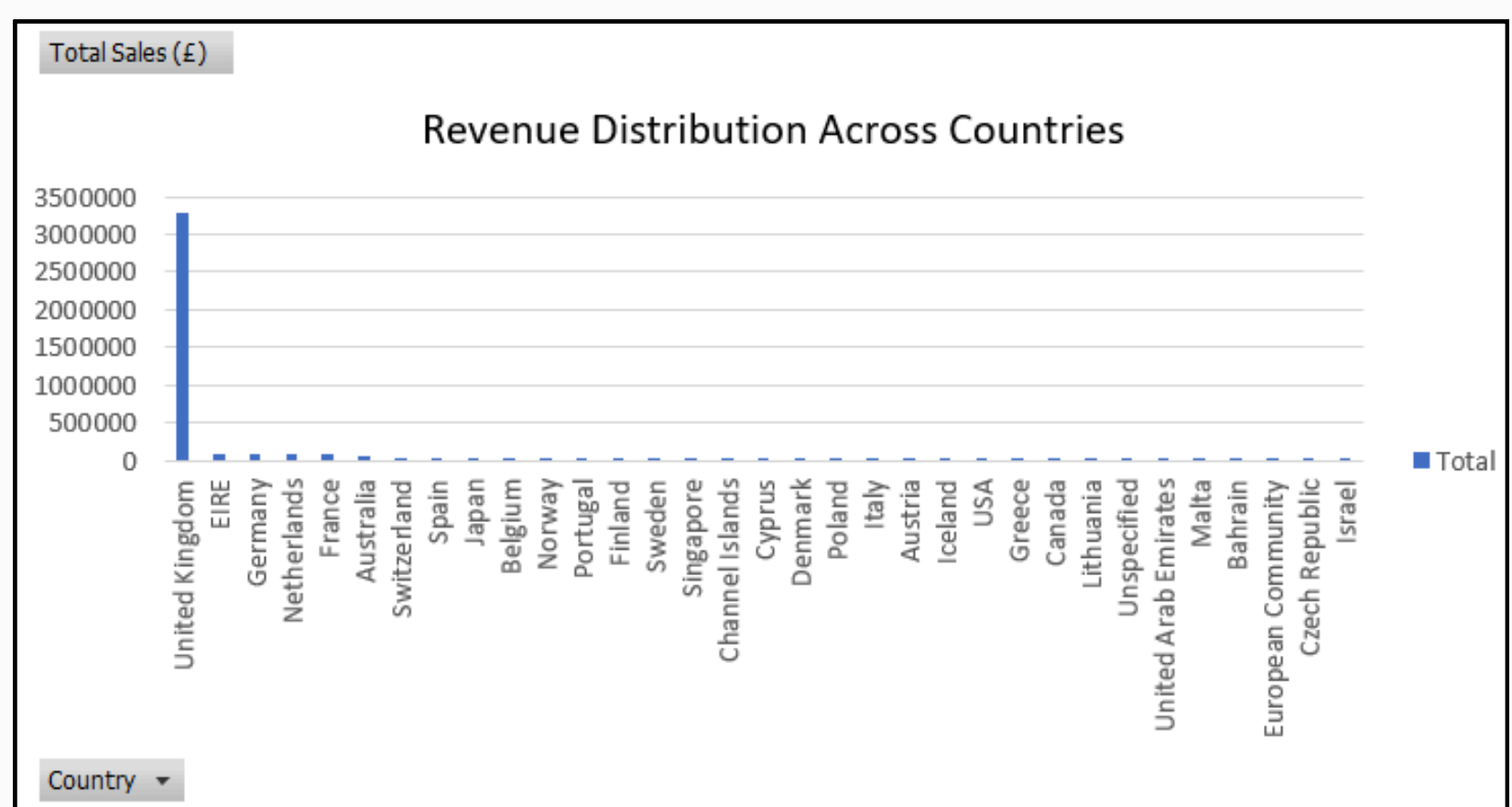


Fig 8. Column Chart for Revenue Distribution Across Countries

Summary

The EDA confirms:

- High transaction volume and strong seasonality
- A skew toward low-cost, fast-moving giftware items
- Heavy reliance on the UK market
- A substantial proportion of missing CustomerIDs, requiring careful data preparation
- Clear behavioural patterns supporting downstream segmentation analysis

5. Methodology

The analytical workflow followed a structured, end-to-end methodology designed to ensure data integrity, model robustness and business relevance. The process combined best practices from data cleaning, feature engineering and machine learning modelling to deliver reliable customer segments for strategic decision-making.

1. Data Cleaning & Preparation

The raw transaction data required comprehensive pre-processing to correct operational inconsistencies and ensure suitability for customer-level analysis. Key steps included:

- Removal of cancelled orders by excluding all invoice numbers beginning with “C”.
- Elimination of negative quantities and prices, which represented returns or erroneous entries.
- Conversion of timestamps to standardized datetime format.
- Dropping missing Customer IDs, as unique identifiers were essential for customer-level aggregation.
- Creation of a Sales metric ($\text{Quantity} \times \text{UnitPrice}$) to measure transaction-level revenue.
- Filtering infinite or abnormal values generated during calculation of derived metrics.

2. Feature Engineering: RFM Metrics

To understand customer behaviour holistically, transaction-level data was aggregated into RFM (Recency, Frequency, Monetary) features:

- Recency: Days since the customer's most recent purchase, relative to the snapshot date.
- Frequency: Number of distinct purchase invoices per customer.
- Monetary: Total revenue generated by the customer across all valid purchases.
- Average Purchase Value: Monetary value divided by frequency to capture typical spending per transaction.

3. Data Standardization

Because RFM features exist on different scales (e.g., Recency measured in days vs. Monetary in currency), the data was standardized prior to modelling.

A RobustScaler was used to normalize values while minimizing the influence of extreme outliers, a common characteristic in retail datasets.

4. Determining the Optimal Number of Clusters

To select the ideal segmentation granularity, two statistical approaches were applied:

- Elbow Method: Evaluated within-cluster sum of squares (WCSS) across cluster counts ($k = 2$ to 10) to identify diminishing returns in variance reduction.
- Silhouette Score: Measured cluster separation and cohesion to evaluate the quality of clustering at each k .

These techniques jointly informed the choice of an optimal 4-cluster solution, balancing statistical validity and business interpretability.

5. K-Means Clustering

The standardized dataset was modelled using the K-Means algorithm:

- The model was fit using $k = 4$.
- Each customer was assigned a cluster label representing their behavioural segment.
- Labels were appended to the cleaned RFM table for further analysis and profiling.
- K-Means was selected for its simplicity, scalability and effectiveness in segmenting structured numeric data such as RFM metrics.

6. Cluster Profiling & Interpretation

Post-modelling, each cluster was profiled to interpret its behavioural meaning:

- Cluster means were calculated across all RFM features.
- Customer counts were assessed to evaluate segment size and commercial significance.
- Patterns in recency, purchase frequency and monetary contribution were analysed to derive insights.

6. Segmentation Results

The customer base was segmented using K-Means clustering applied to standardized RFM features. A series of diagnostic analyses including the Elbow Method, Silhouette Scores and PCA projection were used to validate the number of clusters and interpret the structure of the data.

1. Optimal Number of Clusters

To determine the appropriate segmentation granularity, clustering performance was evaluated for k values ranging from 2 to 10.

- Elbow Method:

The WCSS curve suggested that variance reduction starts to diminish around $k = 3$, indicating a natural levelling point. However, additional evaluation was considered to determine the most interpretable and actionable segmentation.

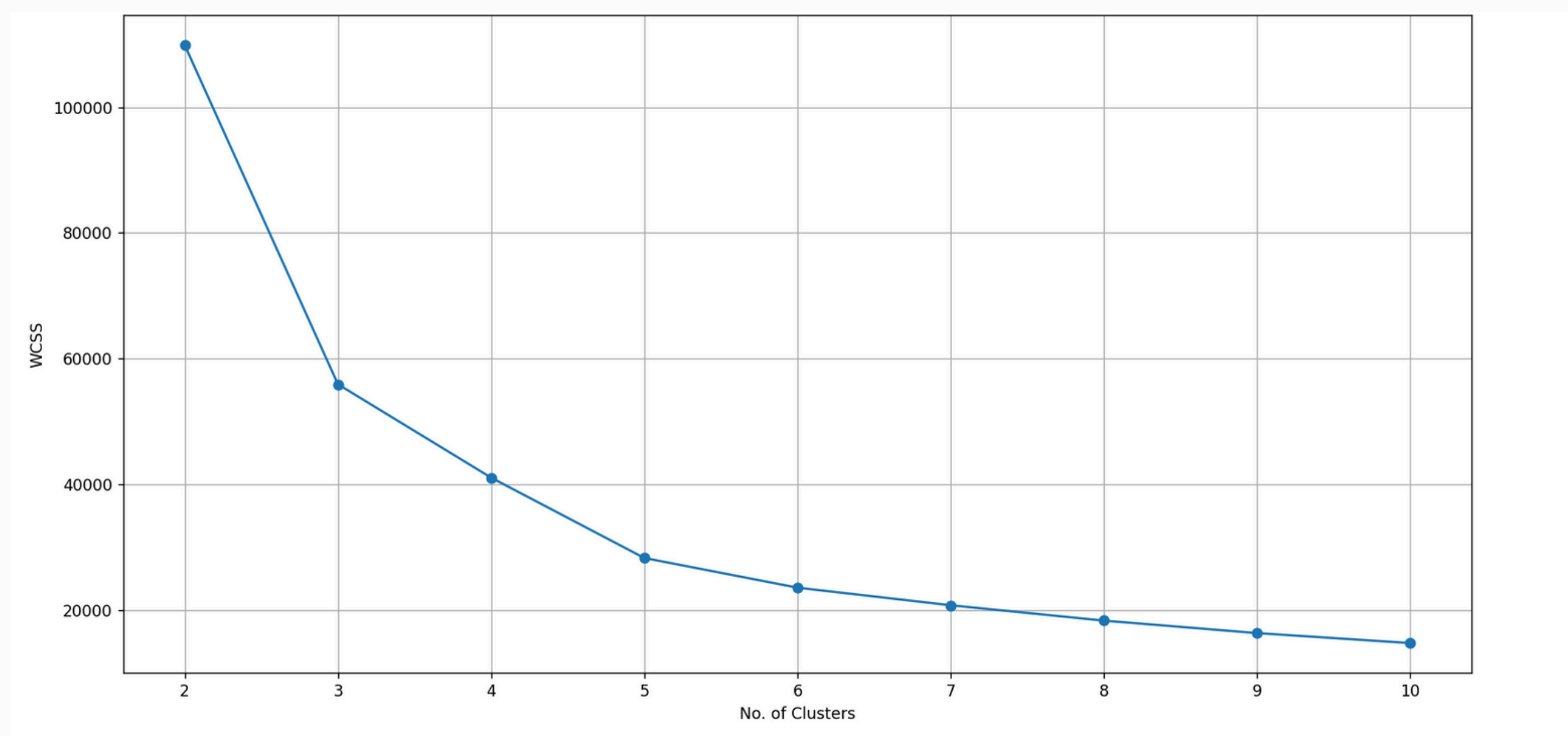


Fig 9. Elbow Method for Optimal K

- Silhouette Score Analysis

Silhouette values reached a peak near $k = 2$, reflecting strong cohesion within a very small number of clusters. However, such a coarse segmentation would provide limited insights into distinct customer behaviours. The score at $k = 4$ remained reasonably high, striking a balance between cluster separation and practical interpretability. Notably, although $k = 3$ exhibited slightly higher silhouette value than $k = 4$, the resulting clusters were less suitable for differentiating meaningful customer segments, as they would create an overly broad grouping in one cluster and reduce business applicability.

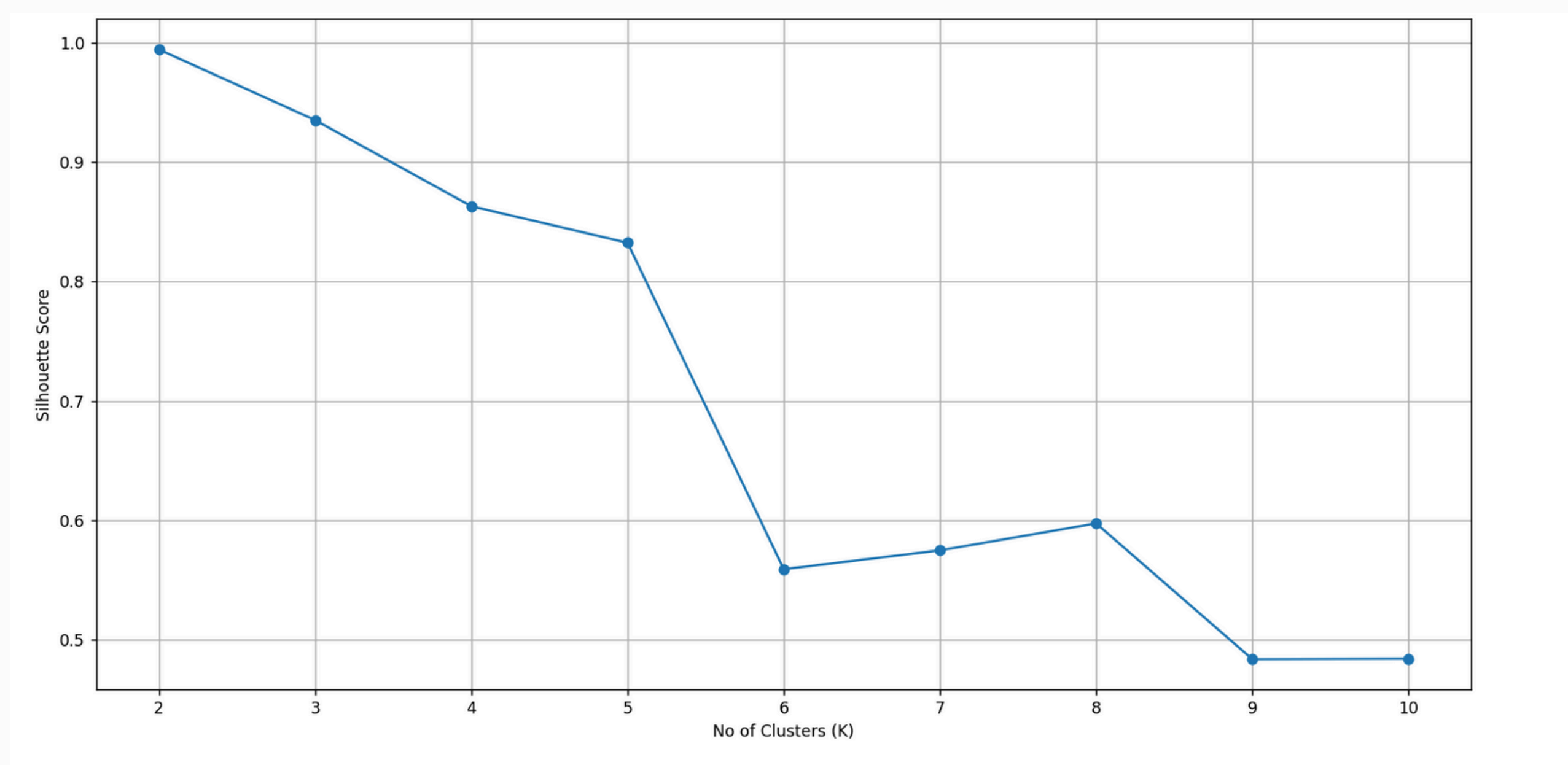


Fig 10. Silhouette Score for each K

Selection of Number of Clusters :

A comparison of the 3-cluster and 4-cluster K-Means models showed that the 4-cluster configuration offers stronger analytical validity and clearer business value. Key reasons include :

i. Better Behavioural Differentiation -

- The 3-cluster model merged mid-value and high-value customers into the same group, masking important differences.
- The 4-cluster model separates:
 - Low-value / inactive customers
 - Mid-value but active customers
 - High-value loyal buyers
 - A singular extreme outlier
- This results in more meaningful and targeted insights.

ii. Stronger Support from Elbow Method -

- The WCSS curve showed a clear bend at $k = 3$, while $k = 4$ exhibited only a subtle inflection. Despite this, $k = 4$ was considered to allow more meaningful differentiation among customer segments.
- $k = 3$ still showed relatively large unexplained variance.

iii. Higher Silhouette Score Stability -

- The 4-cluster configuration delivered better separation and cohesion.
- The 3-cluster model showed:
 - More overlap between behavioural groups
 - Reduced internal cluster tightness
- This makes 4 clusters statistically more reliable.

iv. More Balanced and Realistic Customer Segmentation -

- In the 3-cluster model, one cluster dominated the dataset ($\approx 2950+$ customers), leading to over-generalized insights.
- The 4-cluster model distributes customers more appropriately across behavioural categories, making it easier to design targeted strategies for each group.

v. Preserves Strategic Granularity -

- With only 3 clusters, mid-tier and premium customers collapse together, limiting personalization and marketing differentiation.
- The 4-cluster solution maintains these distinctions, providing greater strategic clarity.

Based on clustering validity, behavioural logic and strategic usefulness, the 4-cluster model is clearly superior and was selected as the final segmentation structure.

Cluster	Recency	Frequency	Monetary	AveragePurchase	Count
0	156.420539	2.460606	930.515822	368.515199	2970
1	90.000000	1.000000	168469.600000	168469.600000	1
2	39.884615	25.076923	36284.092308	2979.387090	26

Fig 11. 3-Cluster Segmentation

Cluster	Recency	Frequency	Monetary	AveragePurchase	Count
0	157.452779	2.327651	814.727903	352.895392	2933
1	90.000000	1.000000	168469.600000	168469.600000	1
2	63.019231	17.173077	13592.026346	1704.557903	52
3	47.272727	21.818182	55512.370909	4388.637411	11

Fig 12. 4-Cluster Segmentation

2. PCA Visualization

A two-component PCA projection was generated to visualize the high-dimensional RFM data in a 2D space.

- The four clusters formed clear, distinguishable groups in the PCA plot.
- Cluster centroids appeared well-separated, reinforcing that the K-Means algorithm successfully captured meaningful behavioural segments.
- Dense and compact groupings of data points indicated coherent internal structure within clusters.

Although PCA reduces dimensionality, the visual separation provides qualitative assurance that the four-cluster model aligns well with the underlying data distribution.

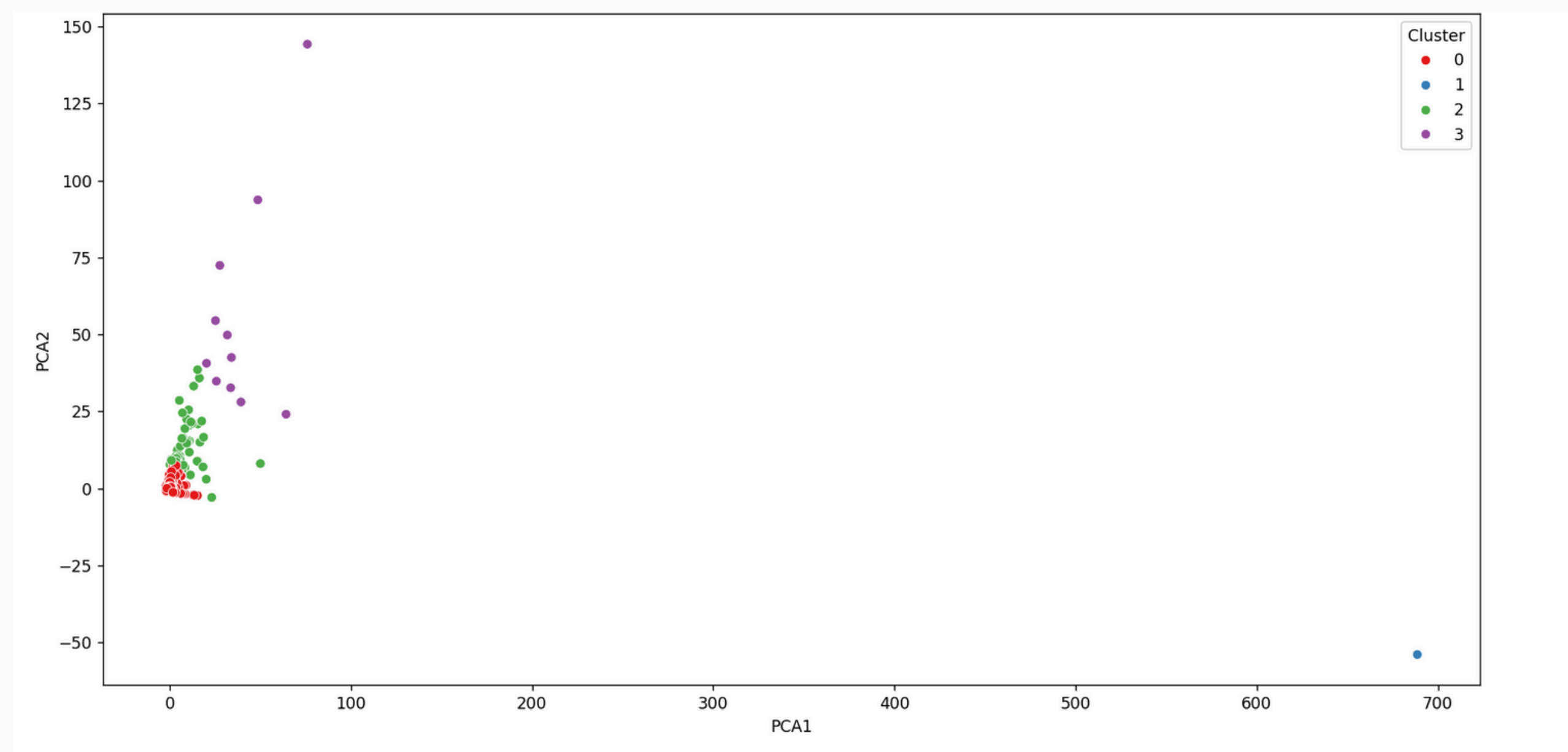


Fig 13. 2-D PCA Visualization for Customer Segments

3. Cluster Profiles

The final cluster characteristics, based on unscaled RFM metrics, are summarized below:

Cluster	Recency (↓ recent)	Frequency (↑ loyal)	Monetary (↑ high-value)	Avg. Purchase	Count
0	157 days	2.33	£815	£353	2933 Customers
1	90 days	1.00	£168,470	£168,470	1 Customer
2	63 days	17.17	£13,592	£1,705	52 Customers
3	47 days	21.82	£55,512	£4,983	11 Customers

Fig 14. Cluster-wise Feature Engineering

Cluster Interpretation

Cluster 0 – Low-Frequency, Low-Value, Long-Inactive (Majority Group)

- Largest group (≈ 2933 customers)
- High Recency → purchased long ago
- Very low Frequency
- Minimal Monetary value
- Represents the core base of disengaged or one-time shoppers

Cluster 1 – Exceptional Outlier (Ultra-High Spender)

- Single customer
- Extremely high Monetary and Average Purchase values
- Not suitable for general strategy; treated as a special or VIP case

Cluster 2 – Active Repeat Buyers with Moderate Value

- Recency relatively low (active users)
- High Frequency (~17 invoices)
- Moderate Monetary value
- Represents consistent mid-value loyal customers, likely frequent small-basket purchasers

Cluster 3 – Highly Loyal, High-Value Customers (Power Users)

- Lowest Recency → very recent purchases
- Very high Frequency (~22 invoices)
- High Monetary value
- Small in number but strategically critical
- Represents premium customers with strong revenue contribution

4. Overall Segmentation Insight

The four-cluster solution provides a balanced and actionable segmentation structure:

- A large reactivation segment (Cluster 0) that requires engagement strategies.
- A small but high-value VIP segment (Clusters 2 & 3) that drives a disproportionate share of revenue.
- A true outlier segment (Cluster 1) that should be monitored separately.
- Clear behavioural distinctions that can inform targeted marketing, retention planning and personalized communication.

7. Business Insights and Interpretation

The four-cluster segmentation model uncovers clear behavioural patterns within the customer base, enabling targeted commercial strategies. Each cluster differs in purchasing frequency, monetary value and recent activity, providing a structured understanding of customer heterogeneity.

1. **Cluster 0** — Low-Frequency, Low-Value, Inactive Majority (≈2933 customers)

This is the largest customer segment, characterized by high recency, low purchase frequency and modest spending. Customers in this group typically made infrequent or one-time purchases and have not engaged with the business for a considerable period.

Interpretation -

- Represents customers with weak loyalty and high attrition risk.
- Marketing influence on this segment is generally low unless reactivation campaigns are deployed.
- Their low contribution to revenue suggests they should not be prioritized for resource-intensive initiatives.

2. **Cluster 1** — Single High-Value Outlier (1 customer)

This cluster contains a single extremely high-spending customer with exceptionally large monetary value and average purchase size.

Interpretation -

- This is a statistical outlier rather than a strategic segment.
- It reflects the presence of an unusually large transaction that does not align with typical customer behavior.
- Should not drive strategic decisions but highlights the importance of occasionally handling high-value bulk buyers differently (e.g., B2B purchases).

3. **Cluster 2** — Mid-Value, Moderately Active Customers (~52 customers)

Customers in this segment show moderate recency and significantly higher purchase frequency compared to Cluster 0. Their monetary contribution is substantial but not extreme.

Interpretation -

- These are engaged retail buyers who purchase relatively frequently and maintain a consistent relationship.
- They represent a valuable segment with strong potential for loyalty-building and upsell opportunities.
- High responsiveness to targeted promotions and product recommendations is expected.

4. **Cluster 3** — High-Value, Highly Active Loyal Customers (~11 customers)

This is the segment with the best overall customer health with low recency, high frequency and very strong monetary value. These customers buy often, recently and generate outsized revenue relative to their size.

Interpretation -

- This group represents the core of the customer base and is critical for revenue stability.
- They should be prioritized for loyalty programs, exclusive offers and long-term retention strategies.
- High repeat engagement and high purchase volume indicate strong alignment with the product assortment.

Cross-Cluster Business Interpretation

1. Revenue Concentration Among High-Value Customers

Despite being a small segment, high-value customers (Clusters 2 and 3) generate a disproportionate share of revenue. Targeted retention and premium experiences for these segments can directly influence profitability.

2. Majority of Customers Are Dormant or Low-Value

Cluster 0 dominates the customer base, indicating a long tail of occasional buyers. Improving retention among this group, even marginally, can significantly expand recurring revenue.

3. Clear Differentiation Between Active and Inactive Customers

Recency patterns strongly highlight two behavioural blocks:

- Active, recently engaged customers: Clusters 2 and 3
- Inactive, low-engagement customers: Cluster 0
- This split helps define where personalized engagement can deliver the highest ROI.

4. Identification of an Outlier Buyer

The existence of a single extreme outlier (Cluster 1) underlines the importance of monitoring abnormal purchasing patterns and separating them from core customer behaviour during analysis.

8. Strategic Recommendations

Based on the insights derived from the four customer segments, the following strategic actions are recommended to maximize customer value, improve retention and drive revenue growth:

1. High-Value Loyal Customers (Cluster 3)

- Priority: Highest
- Strategy:
 - Implement exclusive loyalty programs, premium memberships or early-access benefits.
 - Offer high-touch service, personalized product recommendations and dedicated support lines.
 - Introduce referral incentives to encourage advocacy among this already-engaged customer base.
- Expected Impact: Increased lifetime value and brand loyalty.

2. High-Spend but Low-Frequency Customers (Cluster 1)

- Priority: High
- Strategy:
 - Launch targeted reactivation campaigns to increase purchase frequency (e.g., reminders, product bundles, season-based promotions).
 - Provide personalized high-value offers, such as VIP discounts or curated product selections.
 - Explore subscription-based offerings suitable for their spending patterns.
- Expected Impact: Significant revenue uplift by converting high spenders into repeat purchasers.

3. Moderate-Value, Growth-Potential Customers (Cluster 2)

- Priority: Medium
- Strategy:
 - Encourage higher purchases through tiered loyalty benefits, volume discounts or “spend more, save more” campaigns.
 - Offer cross-sell and upsell recommendations driven by their moderate recency and relatively high frequency.
 - Use behaviour-based promotions to gradually move them into the high-value category.
- Expected Impact: Gradual increase in Average Revenue Per User (ARPU) and overall customer value.

4. At-Risk or Low-Engagement Customers (Cluster 0)

- Priority: Moderate
- Strategy:
 - Implement recency-based win-back campaigns, including limited-time discounts or personalized reminders.
 - Improve engagement by sharing educational content, product use cases or value-based messaging.
 - Optimize marketing cost by applying cost-efficient channels like email and SMS rather than high-cost campaigns.
- Expected Impact: Controlled budget spending and improved reactivation rates.

5. Portfolio-Wide Recommendations

- Customer-Led Personalization: Deploy targeted messaging based on each segment's behavioural traits to enhance relevance.
- Product Strategy Alignment: Adjust product bundles, pricing and offers to align with the needs of different segments.
- Retention Focus: Prioritize reducing churn within low-engagement segments while cultivating loyalty in high-value groups.
- Monitoring Framework: Establish KPIs (CLV, repeat purchase rate, segment migration rate) to measure progress and refine strategies.

9. Implementation Plan

1. Phase-Wise Rollout

Phase 1 : Setup (Weeks 1-2)

- Add final cluster labels to the customer database.
- Share segmentation profiles across teams.
- Establish baseline KPIs (repeat rate, AOV, churn).

Phase 2 : Execution (Weeks 3-6)

- Launch targeted campaigns for each segment:
 - Loyalists: retention & loyalty incentives
 - High-Potential: upsell/cross-sell offers
 - Price-Sensitive: promotional outreach
 - At-Risk: reactivation and win-back messaging
- Configure automated CRM workflows.

Phase 3 : Optimization (Weeks 7-12)

- Track weekly campaign performance.
- A/B test offers and refine messaging.

2. Ownership & Tools

Teams & Roles:

- Marketing: segment-specific campaigns
- Sales: high-value customer engagement
- CRM: workflow automation & customer journeys
- Data Team: maintain segmentation logic, dashboards

3. KPIs for Success

- Repeat purchase rate
- Frequency & monetary uplift
- Campaign conversion rate
- Churn reduction for at-risk customers

4. Review Cadence

- Quarterly: update RFM metrics & cluster labels
- Bi-Annual: review segmentation model performance

5. Expected Impact

- Higher customer lifetime value
- Better targeted spend and campaign efficiency
- Improved retention and personalization

10. Conclusion

This customer segmentation analysis provides a structured and data-driven understanding of customer behaviour, purchasing patterns and value contribution. By leveraging RFM metrics and K-Means clustering, the study successfully identified four distinct customer segments, each representing unique engagement levels and business opportunities. These segments form the foundation for actionable strategies across retention, acquisition, personalization and revenue optimization.

The insights derived from this analysis empower the organization to shift from broad, uniform outreach to precise, segment-specific decision-making. When integrated into CRM workflows and marketing operations, this segmentation enables more relevant customer experiences, improved efficiency in resource allocation and measurable uplift in customer lifetime value.